



# Modelling Financial Markets

## Chapter 4: Portfolio Optimisation



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## Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LO: Global Minimum-Variance portfolio, Long-Only
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit–Wolf covariance
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ HYG: iShares iBoxx High Yield Corporate Bond ETF
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ LO: Long-Only, no short sales
- ▣ LQD: iShares iBoxx Investment Grade Corporate Bond ETF
- ▣ MV: Mean–Variance tangency portfolio estimated from sample moments
- ▣ MV-LO: Mean–Variance tangency portfolio, Long-Only
- ▣ OLS: Ordinary Least Squares
- ▣ OOS: Out-Of-Sample
- ▣ POET: Principal Orthogonal complement Thresholding
- ▣ US: United States

## Outline

Motivation

Mean–Variance Optimisation

The Limits of Markowitz

Covariance Shrinkage

Black–Litterman

Risk Parity and Hierarchical Risk Parity

Out-of-Sample Evaluation

A Portfolio of BVB Blue Chips

AI for Scientific Discovery

Conclusions

References

## Portfolio Optimisation: Easy to Write, Hard to Estimate

- A **portfolio** is a weight vector  $w = (w_1, \dots, w_N)^\top$  with  $w^\top 1 = 1$ ; optimisation chooses  $w$  to maximise a criterion under constraints
- The inputs are **estimated**:  $N$  means and  $N(N+1)/2$  variances and covariances
  - ▶ 16 assets:  $16 + 136 = 152$  parameters from a 60-month window, i.e. 960 observations
- Uses: strategic allocation (pension funds), home bias of private investors, the covariance matrix of risk management (Chapters 6–7)
- Key message: judge every rule **out of sample**, after costs, with a test of the difference
- Textbook treatment: [Franke, Härdle & Hafner \(2019\)](#), Part I (portfolio theory)

## The Central Tension

### In theory

- ▣ Known means and covariances
- ▣ The mean–variance portfolio is optimal
- ▣ More assets always help

### In practice

- ▣ Parameters are estimated with error
- ▣ The optimiser loads on the largest errors
- ▣ More assets, more parameters, more noise

- ▣ This chapter: better inputs (shrinkage, Black–Litterman), rules that ignore means (risk parity, HRP) and a fair out-of-sample comparison

## Data Used in This Chapter

| Universe    | Assets                                  | Period    |
|-------------|-----------------------------------------|-----------|
| Sectors     | 9 US sector ETFs (XLB ... XLY)          | 1999–2026 |
| Multi-asset | SPY, EFA, EEM, TLT, IEF, LQD, HYG, GLD  | 2007–2026 |
| Combined    | 9 sectors + 7 multi-asset ETFs (no SPY) | 2007–2026 |
| Romania     | 8 BVB blue chips, BET, BET-TR           | 2014–2026 |

- Monthly returns from month-end prices on common trading days; US data end in July 2026, BVB data in August 2026
- Risk-free rate: one-month Treasury bill, [Kenneth French Data Library](#)

## The Assets

- ▣ Equities: SPY (S&P 500), EFA (developed markets outside North America), EEM (emerging markets)
- ▣ US Treasuries: TLT (maturities above 20 years), IEF (7–10 years)
- ▣ Credit: LQD (investment-grade corporate bonds), HYG (high-yield corporate bonds)
- ▣ Gold: GLD; sectors: XLK technology, XLU utilities, XLP consumer staples, . . .
- ▣ BVB: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL; BET is a price index, BET-TR reinvests dividends

## Prerequisites Recap: Notation and Two Assets

- Simple return  $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$ ; excess return  $R_{i,t}^e = R_{i,t} - R_{f,t}$ ; portfolio mean  $w^\top \mu$ , variance  $w^\top \Sigma w$
- **Sharpe ratio** (Sharpe (1966)):  $SR_p = \mathbb{E}[R_p^e]/\sigma(R_p^e)$ ; monthly figures annualised with  $\times 12$  (mean) and  $\times \sqrt{12}$  (volatility, Sharpe)
- **Maximum drawdown**:  $\min_t (W_t / \max_{s \leq t} W_s - 1)$ , the largest fall of wealth  $W_t$  from a previous peak
- Two assets, correlation  $\rho$ :  $\sigma_p^2 = w^2 \sigma_1^2 + (1-w)^2 \sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$ , minimised at  $w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
- SPY and TLT, monthly 2002–2026:  $\rho = -0.10$ ,  $w^* = 46\%$  SPY, volatility 9.4% against 14.7% and 13.5% (time-varying correlation: Chapter 6)

## The Markowitz Problem

- ▣ **Markowitz (1952)**: investors care about the mean and the variance of portfolio returns
- ▣ **Minimum-variance frontier**: minimum variance for every target mean  $m$ ; its upper branch,  $m \geq \mu_{\text{GMV}}$  (the mean of the GMV portfolio), is the **efficient frontier**

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = m, \quad w^\top \mathbf{1} = 1$$

- ▣ Utility form, risk aversion  $\gamma > 0$ :  $\max_w w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w$  s.t.  $w^\top \mathbf{1} = 1$ ; as  $\gamma$  varies it traces exactly the efficient branch
- ▣ Inputs:  $N$  means,  $N$  variances,  $N(N-1)/2$  covariances
- ▣ Weights may be negative: a **short sale** (selling a borrowed asset)

## Deriving the GMV and Tangency Portfolios for $N$ Assets

- GMV:  $\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$  s.t.  $\mathbf{w}^\top \mathbf{1} = 1$ ; Lagrangian  $\mathcal{L} = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{1} - 1)$ 
  - ▶ First-order condition  $2\Sigma \mathbf{w} = \lambda \mathbf{1}$ , so  $\mathbf{w} = \frac{\lambda}{2} \Sigma^{-1} \mathbf{1}$ ; the constraint gives  $\frac{\lambda}{2} = 1/A$ ,  $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$
  - ▶  $\mathbf{w}_{\text{GMV}} = \Sigma^{-1} \mathbf{1}/A$  and  $\sigma_{\text{GMV}}^2 = \mathbf{w}_{\text{GMV}}^\top \Sigma \mathbf{w}_{\text{GMV}} = 1/A$
- $N = 2$ :  $\Sigma^{-1} \propto \begin{pmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{pmatrix}$  returns  $\mathbf{w}^*$  of the recap slide
- Tangency: maximise  $\mathbf{w}^\top \boldsymbol{\mu} / \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$ ; the condition  $\boldsymbol{\mu} \propto \Sigma \mathbf{w}$  gives  $\mathbf{w} \propto \Sigma^{-1} \boldsymbol{\mu}$
- Maximum squared Sharpe ratio:  $\theta^2 = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$ , the quantity estimated with bias in Section 3
- With  $\Sigma$  positive definite and  $B = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$ , the normalised weights  $\Sigma^{-1} \boldsymbol{\mu} / B$  have Sharpe ratio  $\text{sign}(B) \theta$ 
  - ▶ Maximum only if  $B > 0$ ;  $B < 0$  gives the minimum,  $B = 0$  no fully invested solution

## Tobin: Separation



James Tobin, Nobel Prize in Economics 1981

Photo: unknown author (1962); public domain;  
[Wikimedia Commons](#)

- **Tobin (1958)**: add a riskless asset paying  $R_f$
- Every investor holds the **same** risky portfolio, mixed with cash or borrowing
- Risk aversion changes only the mix, not the risky portfolio: **two-fund separation**
- The best risky portfolio maximises the Sharpe ratio: the **tangency portfolio**
- Separation is the bridge from Markowitz to the CAPM (Chapter 3)

## Two Special Portfolios

### Global minimum variance (GMV)

- Lowest variance of all portfolios

$$w_{\text{GMV}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}$$

- Needs **no** expected returns

- In this chapter, MV denotes the tangency rule estimated from sample means and covariances

### Tangency (maximum Sharpe)

- Highest Sharpe ratio

$$w_{\text{MV}} = \frac{\Sigma^{-1} \mu}{\mathbf{1}^\top \Sigma^{-1} \mu}$$

- $\mu$ : expected **excess** returns; requires  $\mathbf{1}^\top \Sigma^{-1} \mu > 0$

## Merton: the Frontier in Closed Form



Robert C. Merton, Nobel Prize in  
Economics 1997

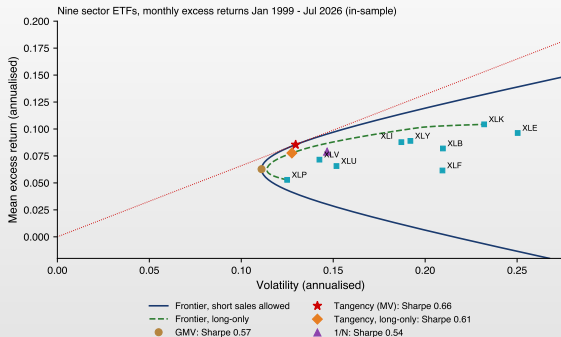
Photo: Digarnick (2006); public domain;  
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- Merton (1972): with short sales, the frontier is a hyperbola
- Let  $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$ ,  $B = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$ ,  $C = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$ ,  $D = AC - B^2$

$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$

- The GMV sits at  $m = B/A$  with  $\sigma^2 = 1/A$
- Every frontier portfolio is a mix of GMV and tangency

## The Frontier of US Sectors



- Monthly excess returns, January 1999 – July 2026 ( $T = 331$ ); all parameters estimated on the full sample

## Reading the Sector Frontier

- ▣ Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%);  $1/N$ : 0.54
- ▣ Tangency weights:  $-0.37$  financials,  $-0.21$  materials,  $+0.37$  health care
- ▣ Long-only tangency: Sharpe 0.61; materials, financials and industrials get exactly zero
  - ▶ A **corner solution**: the constraint binds and assets drop out
- ▣ The gain of the optimiser over  $1/N$  is small even **in sample**, with perfect hindsight

## Constraints

- ▣ **Long-only (LO)**:  $w_i \geq 0$ ; no closed form, solved by quadratic programming
- ▣ Position limits:  $w_i \leq 0.2$ ; sector or asset-class limits:  $\sum_{i \in G} w_i \leq c_G$
- ▣ Leverage limit:  $\sum_i |w_i| \leq L$
- ▣ **Frost & Savarino (1988)**: constraining weights improved out-of-sample performance
- ▣ In this chapter: MV-LO and GMV-LO are the long-only versions of MV and GMV

## Recap: Mean–Variance Optimisation

- ▣ The frontier trades mean against variance; correlation drives diversification
- ▣ Two special portfolios: GMV, which needs no means, and tangency, which maximises the Sharpe ratio
- ▣ With a riskless asset, the risky part of every efficient portfolio is the tangency portfolio
- ▣ Long-only constraints create corner solutions
- ▣ Everything so far assumed the parameters are **known**

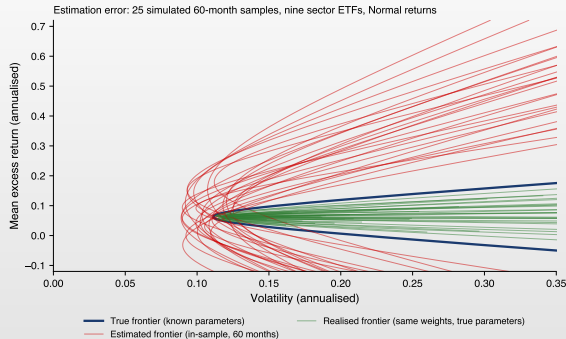
## Means Are Hard to Estimate: Merton's Argument

- Merton (1980):  $Y$  years sampled  $h$  times a year, i.i.d. returns with annual mean  $\mu$  and variance  $\sigma^2$ 
  - ▶  $\text{Var}(\hat{\mu}) = \sigma^2/Y$ : independent of  $h$ ; only the **span** of the sample helps
  - ▶  $\text{Var}(\hat{\sigma}^2) \approx 2\sigma^4/(hY) \rightarrow 0$  as  $h \rightarrow \infty$ : variances sharpen with frequency
- SPY,  $Y = 24$ ,  $\sigma = 14.7\%$ : standard error 3.0%, 95% interval [5.8%, 17.6%] for a mean of 11.7%
- A standard error of 1% needs  $(0.147/0.01)^2 \approx 217$  years
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances; rules without means (GMV, risk parity) are more robust

## Error Maximisation

- Michaud (1989): mean–variance optimisers are **estimation-error maximisers**
  - ▶ They overweight assets with overestimated means and underestimated risk
- Best & Grauer (1991): tiny changes in a mean can move the weights of many assets
- In-sample performance is therefore **biased upwards**
- Experiment: simulate returns from known parameters, optimise, evaluate with the true parameters

## Estimated versus Realised Frontiers



- True parameters: sector sample means and covariances, 1999–2026; returns drawn from the Normal distribution

## Reading the Simulation

- ▣ Red: frontiers estimated from 60 months, far too optimistic
- ▣ Green: the same portfolios evaluated with the true parameters, all **below** the true frontier
- ▣ 2000 simulated tangency portfolios, 60 months each
  - ▶ Median estimated maximum Sharpe  $\hat{\theta}$  1.56; median true Sharpe 0.30; true optimum 0.66
  - ▶ In 11% of the samples  $1^\top \hat{\Sigma}^{-1} \hat{\mu} < 0$ : the normalised weights are then the in-sample minimum
  - ▶ 96% have a true Sharpe below  $1/N$  (0.54); 7.5% below zero
- ▣ Even with Normal, stable returns, estimation error destroys the optimum

## The Plug-in Sharpe Ratio Is Biased: an Exact Result

- i.i.d.  $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  returns, maximum-likelihood  $\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}$ :  $\hat{\boldsymbol{\mu}} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}/T)$ ,  $T\hat{\boldsymbol{\Sigma}} \sim W_N(T-1, \boldsymbol{\Sigma})$  (Wishart), independent
- Inverse Wishart moment, finite for  $T > N + 2$ :  $\mathbb{E}[\hat{\boldsymbol{\Sigma}}^{-1}] = \frac{T}{T-N-2}\boldsymbol{\Sigma}^{-1}$ , an **upward** bias that grows with  $N/T$
- Hence, with  $\hat{\theta}^2 = \hat{\boldsymbol{\mu}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$  (Kan & Zhou (2007))

$$\mathbb{E}[\hat{\theta}^2] = \frac{T}{T-N-2} \left( \theta^2 + \frac{N}{T} \right) = \frac{T\theta^2 + N}{T-N-2}$$

- Sectors:  $N = 9$ ,  $T = 60$ , true  $\theta = 0.66$  a year:  $\mathbb{E}[\hat{\theta}^2]$  corresponds to a Sharpe of 1.65 a year
  - ▶ The analytical counterpart of the simulated median of 1.56 on the previous slide

## Correcting the Bias and the Distribution of Weights

- ▣ Unbiased estimator of  $\theta^2$  (Kan & Zhou (2007)):  $\hat{\theta}_u^2 = [(T - N - 2)\hat{\theta}^2 - N]/T$ 
  - It can be negative (then truncated at zero), and  $\sqrt{\hat{\theta}_u^2}$  is not unbiased for  $\theta$
- ▣ Sectors, full sample ( $T = 331$ ,  $N = 9$ ): plug-in  $\hat{\theta} = 0.66$ , bias-corrected  $\sqrt{\hat{\theta}_u^2} = 0.31$  a year
  - The correction removes more than half of the in-sample tangency Sharpe, even with 27 years of data
- ▣ GMV weights:  $\hat{w}_{\text{GMV}}$  is unbiased and its exact distribution and covariance are known (Okhrin & Schmid (2006))
- ▣ Tangency weights divide by  $1^\top \hat{\Sigma}^{-1} \hat{\mu}$ , which can be near zero: very heavy tails, the explosions of the rolling weights
- ▣ Estimation-risk-optimal mixtures of tangency, GMV and 1/N: Kan & Zhou (2007), Kan, Wang & Zhou (2022)

## Tangency Weights by Regression: Britten-Jones (1999)

- Britten-Jones (1999): regress the constant 1 on the excess returns, **without** a constant term

$$1 = \mathbf{b}^\top \mathbf{r}_t + u_t, \quad \hat{\mathbf{b}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{1} \propto \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}$$

- Sherman–Morrison on  $\mathbf{X}^\top \mathbf{X} / T = \hat{\Sigma} + \hat{\boldsymbol{\mu}} \hat{\boldsymbol{\mu}}^\top$  gives the proportionality (Seminar 4, A2);  $\hat{\mathbf{b}} / \mathbf{1}^\top \hat{\mathbf{b}} =$  sample tangency weights
- OLS  $t$ -test of  $b_i = 0$ : asset  $i$  has zero weight in the tangency portfolio
- $F$ -test of  $\mathbf{R}\mathbf{b} = 0$ : for example  $b_1 = \dots = b_N$ , i.e. “the tangency portfolio is  $1/N$ ”
- Exact finite-sample  $t$  and  $F$  distributions under i.i.d. Normal returns; HAC standard errors otherwise

## Britten-Jones on the Nine Sectors

- Monthly excess returns, January 1999 – July 2026 ( $T = 331$ ); normalised  $\hat{b}$  reproduces the tangency weights ( $-0.37$  financials,  $+0.37$  health care)
- No single weight differs from zero: largest  $|t| = 1.12$  (financials); with HAC errors 1.19
- $H_0$ : tangency =  $1/N$ , eight restrictions  $b_i = b_9$ 
  - ▶  $F(8, 322) = 0.48$ ,  $p = 0.87$ ; HAC Wald  $\chi^2_8 = 4.50$ ,  $p = 0.81$
- 27 years of data cannot tell the estimated optimum from equal weights
- The same verdict as the window-length simulation and the out-of-sample tests below

## Unstable Weights



- Nine sector ETFs; each month the weights use only the previous 60 months (top panel clipped at  $\pm 6$ )

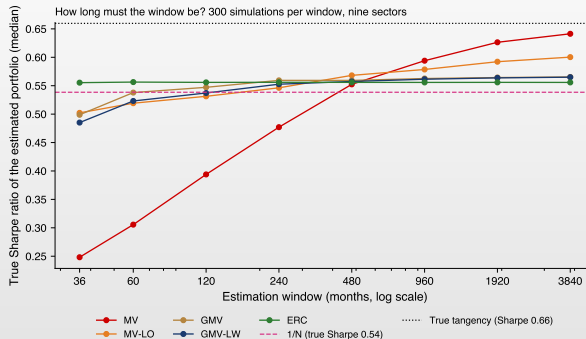
## Reading the Rolling Weights

- ▣ MV: median gross exposure  $\sum_i |w_i| = 4.9$ ; largest single weight 26 in absolute value
- ▣ Gross exposure above 3 in 90% of the months
- ▣ Explosions in 2004–2006 and 2009:  $\mathbf{1}^\top \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}$  close to zero in the denominator
  - ▶ Negative in 12 of 271 windows: MV, as published, then holds the minimum-Sharpe portfolio
- ▣ GMV with shrinkage (next section): gross exposure 1.5, largest weight 0.94
- ▣ Unstable weights mean high turnover, hence high trading costs

## The 1/N Benchmark

- ▣ **1/N**: equal weights  $w_i = 1/N$ , rebalanced each month; no estimation at all
- ▣ DeMiguel, Garlappi & Uppal (2009): 14 optimisation models on 7 data sets
  - ▶ None beats 1/N consistently in Sharpe ratio, certainty-equivalent return or turnover
- ▣ Their calibration: about 3000 months of data for sample MV to beat 1/N with 25 assets
- ▣ Kan & Zhou (2007): combining tangency and GMV reduces estimation risk
- ▣ Question: how long must the window be for our nine sectors?

## How Long Must the Window Be?



- 300 simulated samples per window length; true parameters as in the previous simulation

## Reading the Window-Length Experiment

- ▣ MV beats 1/N only with 480 months (40 years) of data: median true Sharpe 0.55 vs 0.54
- ▣ With 60 months, MV reaches 0.31, less than 60% of 1/N
- ▣ Long-only MV, GMV and GMV with shrinkage are close to 1/N from 60 months on
- ▣ ERC (Section 6) is almost insensitive to the window: 0.56 throughout
- ▣ Here 1/N is nearly efficient: the true tangency Sharpe is only 0.66

## Recap: The Limits of Markowitz

- ▣ Means are estimated with large errors; the optimiser amplifies them
- ▣ In-sample Sharpe ratios of optimised portfolios are biased upwards:  $\mathbb{E}[\hat{\theta}^2] = (T\theta^2 + N)/(T - N - 2)$
- ▣ Britten-Jones regression: tangency weights are OLS coefficients normalised to sum to one; on sectors they are not statistically distinguishable from  $1/N$
- ▣ Sample MV weights are unstable and extreme
- ▣  $1/N$  is a hard benchmark: MV needs decades of data to beat it
- ▣ Remedies: better covariance estimates, better means, or rules that avoid means

## Why the Sample Covariance Matrix Fails

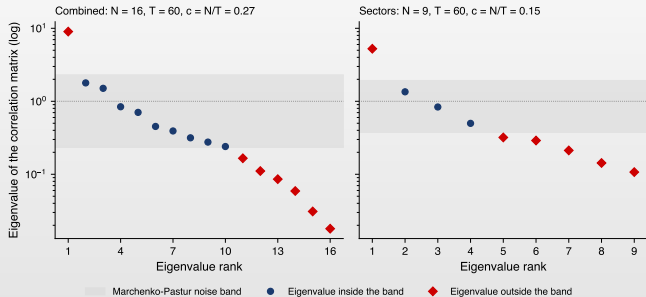
- ▣  $N$  assets:  $N(N + 1)/2$  variances and covariances; 16 assets: 136 parameters from 60 months
- ▣ Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- ▣ **Condition number**: largest over smallest eigenvalue; a large value means  $\Sigma^{-1}$  amplifies noise
- ▣ Optimisers invert  $\Sigma$ , so they load on the directions with the smallest estimated variance
- ▣ The demeaned sample matrix has rank at most  $T - 1$ : singular, not invertible, whenever  $T \leq N$

## Eigenvalues of Pure Noise: Marchenko–Pastur

- i.i.d. returns with  $\Sigma = \sigma^2 I$ : every true eigenvalue equals  $\sigma^2$
- Marčenko & Pastur (1967): as  $N, T \rightarrow \infty$  with  $N/T \rightarrow c \in (0, 1)$ , the sample eigenvalues fill
$$[\lambda_-, \lambda_+] = [\sigma^2(1 - \sqrt{c})^2, \sigma^2(1 + \sqrt{c})^2]$$
- $c = 16/60 = 0.27$ : band  $[0.23, 2.30]$  for a correlation matrix ( $\sigma^2 = 1$ ), a ten-fold spread from noise alone
- Laloux et al. (1999): eigenvalues outside the band are candidates for structure beyond this noise model; those inside cannot be distinguished from it by this comparison alone
- This turns “sample eigenvalues are too dispersed” into a quantitative benchmark

## Sample Eigenvalues versus the Noise Band

Last 60-month window, Aug 2021 - Jul 2026



- Correlation matrices of monthly excess returns, last 60-month window (August 2021 – July 2026); grey band:  $[\lambda_-, \lambda_+]$  for  $\sigma^2 = 1$

## Reading the Eigenvalue Chart

- ▣ 16 ETFs: one eigenvalue above  $\lambda_+ = 2.30$ , the market mode (9.0, 56% of the trace); nine lie inside the band
- ▣ Nine sectors: again only the market mode (5.2) lies above  $\lambda_+ = 1.92$
- ▣ Rescaling the noise variance to  $\sigma^2 = 1 - \lambda_1/N$  (Laloux et al. (1999)): three eigenvalues above the band in each universe
- ▣ Eigenvalues **below** the band (six of 16): near-riskless combinations, e.g. TLT against IEF
  - ▶ GMV loads exactly on these directions: if they are underestimated, its risk is too

## The Shrinkage Estimator

- Shrink the sample matrix  $S$  towards a structured target  $F$

$$\hat{\Sigma}_{\text{LW}} = \delta F + (1 - \delta) S, \quad 0 \leq \delta \leq 1$$

- Targets

- ▶ Single-index (market) model: [Ledoit & Wolf \(2003\)](#)
- ▶ Scaled identity: [Ledoit & Wolf \(2004a\)](#)
- ▶ Constant correlation,  $f_{ii} = s_{ii}$  and  $f_{ij} = \bar{r} \sqrt{s_{ii} s_{jj}}$  for  $i \neq j$ ,  $\bar{r}$  the average sample correlation: [Ledoit & Wolf \(2004b\)](#), used here

- $\delta$  trades the variance of  $S$  against the bias of  $F$

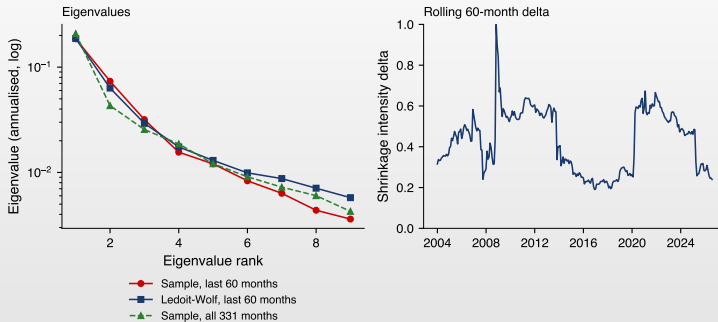
## The Optimal Shrinkage Intensity

- Ledoit–Wolf choose  $\delta$  to minimise the expected squared (Frobenius) distance to the true  $\Sigma$

$$\delta^* = \max \left\{ 0, \min \left\{ 1, \frac{\hat{\kappa}}{T} \right\} \right\}, \quad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$ : sum of the asymptotic variances of the entries of  $S$  (noise)
- $\hat{\rho}$ : their covariances with the target;  $\hat{\gamma} = \|F - S\|^2$  (misspecification)
- Noisy data or a good target: large  $\delta$ ; long samples:  $\delta \rightarrow 0$

## Shrinkage on US Sectors



- Nine sector ETFs, monthly excess returns; last window August 2021 – July 2026

## Reading the Shrinkage Chart

- ▣ Last window:  $\delta = 0.24$ ; the smallest eigenvalues are pulled up, the largest almost unchanged
- ▣ Condition number: sample 51.6, shrunk 32.5
- ▣ Rolling  $\delta$ : mean 0.43, range 0.19–1.00
  - ▶  $\delta = 1$  in the early windows: 60 months are too few to beat the target
- ▣ GMV-LW: the GMV portfolio computed with  $\hat{\Sigma}_{LW}$

## Constraints Are Shrinkage Too

- ▣ Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix

$$\tilde{S} = S - (\lambda \mathbf{1}^\top + \mathbf{1} \lambda^\top), \quad \lambda \geq 0 \text{ the multipliers of } w_i \geq 0$$

- ▶  $\tilde{s}_{ij} = s_{ij} - \lambda_i - \lambda_j$ : a binding constraint on asset  $i$  lowers all its covariances, which shrinks the large estimates
- ▣ Imposing the “wrong” constraint (no short sales) can reduce out-of-sample risk
- ▣ The effect is largest when covariances are estimated from few observations
- ▣ Practical rule: shrink, constrain, or both, but never invert a raw sample matrix of many assets

## Beyond Linear Shrinkage

- Linear shrinkage moves all eigenvalues towards the target with one  $\delta$ ; **nonlinear** shrinkage corrects each eigenvalue separately
  - ▶ Ledoit & Wolf (2017): tuned for portfolio selection; Ledoit & Wolf (2020): an analytical formula, no optimisation
- Factor structure plus a sparse residual covariance: POET (Fan, Liao & Mincheva (2013)), the link to Chapter 3
- High-dimensional GMV with shrinkage towards a target portfolio: Bodnar, Parolya & Schmid (2018)
- Out of sample, 60-month window, GMV volatility
  - ▶ Sectors: sample 11.90%, linear (constant correlation) 11.80%, nonlinear 11.62%
  - ▶ 16 ETFs: sample 4.06%, linear 4.38%, nonlinear 4.31%: better than the linear target, not than the raw matrix

## Recap: Covariance Shrinkage

- The sample covariance matrix is noisy and ill-conditioned when  $N$  is large relative to  $T$
- Shrinkage mixes it with a structured target; the data choose the weight  $\delta$
- The target matters: constant correlation suits one asset class better than a mix of stocks and bonds
- No-short constraints act as implicit shrinkage
- Marchenko–Pastur gives an asymptotic noise benchmark for the eigenvalues; nonlinear shrinkage treats each eigenvalue separately

## Shrinking the Means: One Family of Estimators

- Bayes–Stein (Jorion (1986)):  $\hat{\mu}_{BS} = (1 - \hat{\phi}) \hat{\mu} + \hat{\phi} \hat{\mu}_{GMV}$ 
  - ▶ Empirical-Bayes shrinkage of every mean towards the mean of the GMV portfolio;  $\hat{\phi}$  from a Stein argument, larger when  $N/T$  is large
- Tu & Zhou (2011): the convex combination of a sophisticated rule and  $1/N$  that minimises the expected out-of-sample loss
- DeMiguel, Garlappi, Nogales & Uppal (2009): norm constraints  $\|w\|_1 \leq \delta$  or  $\|w\|_2 \leq \delta$  nest the long-only GMV, shrinkage GMV and  $1/N$ ; the bridge to the lasso portfolios of Chapter 13
- Black–Litterman (next slides): Bayesian shrinkage of  $\mu$  towards the returns implied by a benchmark

## Black and Litterman: Starting from Equilibrium



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia; public domain; Wikimedia Commons

- Black & Litterman (1992), developed at Goldman Sachs
- Problem: sample means give extreme, unintuitive weights
- Idea: start from the returns that make a **benchmark** portfolio optimal
- Add the investor's **views**, weighted by their confidence
- Result: weights that move away from the benchmark only where views exist

## Step 1: Reverse Optimisation

- ▣ A mean–variance investor with risk aversion  $\delta_{\text{RA}}$  and access to cash holds  $w^* = (\delta_{\text{RA}} \Sigma)^{-1} \mu$  in risky assets and  $1 - \mathbf{1}^\top w^*$  in cash
- ▣ Reverse it: the returns that make the benchmark  $w_b$  optimal

$$\pi = \delta_{\text{RA}} \Sigma w_b$$

- ▣  $\pi$ : **implied** (equilibrium) excess returns; with market weights, the CAPM returns
- ▣ Here: no market capitalisations for the sectors, so  $w_b = 1/N$ ,  $\delta_{\text{RA}} = 2.5$

## Step 2: Views and the Posterior

- $K$  views:  $P\mu = q + \varepsilon$ ,  $\varepsilon \sim N(0, \Omega)$ ; prior  $\mu \sim N(\pi, \tau\Sigma)$
- Posterior mean: a precision-weighted average

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1} [(\tau\Sigma)^{-1} \pi + P^\top \Omega^{-1} q]$$

- Weights:  $w_{BL} = (\delta_{RA}\Sigma)^{-1} \mu_{BL}$
- He & Litterman (2002):  $\tau = 0.05$ ,  $\Omega = \text{diag}(P \tau \Sigma P^\top)$ ; with  $\Sigma$  in the weights, only the assets in the views move (the posterior covariance would also scale all weights by  $1/(1 + \tau)$ )

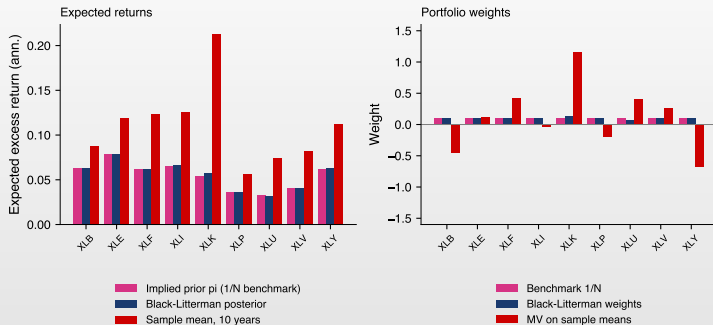
## Black–Litterman as Mixed (GLS) Estimation

- ▣ Theil & Goldberger (1961): stack the prior and the views as one regression on  $\mu$

$$\begin{pmatrix} \pi \\ q \end{pmatrix} = \begin{pmatrix} I \\ P \end{pmatrix} \mu + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}, \quad \text{Var} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{pmatrix} \tau \Sigma & 0 \\ 0 & \Omega \end{pmatrix}$$

- ▣ GLS gives exactly  $\mu_{BL}$  of the previous slide, with covariance  $M = [(\tau \Sigma)^{-1} + P^T \Omega^{-1} P]^{-1}$
- ▣ Equivalent form:  $\mu_{BL} = \pi + \tau \Sigma P^T (P \tau \Sigma P^T + \Omega)^{-1} (q - P \pi)$
- ▣ Limits:  $\Omega \rightarrow 0$  gives  $P \mu_{BL} = q$  exactly (restricted least squares);  $\Omega \rightarrow \infty$  gives  $\pi$
- ▣ One asset,  $\pi = 5\%$ ,  $\tau \sigma^2 = 0.00128$ , view 8% with variance 0.0004:  $\mu_{BL} = 7.29\%$ , view weight 0.76

## Black–Litterman on US Sectors



□  $\Sigma$  from monthly excess returns, August 2016 – July 2026; one view: XLK minus XLU = 3% a year

## Reading Black–Litterman

- ▣ Implied XLK minus XLU: 2.2%; view 3%; posterior 2.6% (the average, since  $\Omega$  equals the prior variance of the view)
- ▣ Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%, all others unchanged
- ▣ Sample means instead: XLK 21.3% a year; MV weights +115% XLK, −68% XLY, −46% XLB
- ▣ The view is relative ( $P1 = 0$ ): the tilt  $\propto P^\top$  sums to zero, so the portfolio stays fully invested; an absolute view would also move the cash position
- ▣ Black–Litterman is Bayesian shrinkage of the means towards equilibrium

## Recap: Black–Litterman

- The prior comes from a benchmark, not from noisy sample means
- Views enter with explicit confidence; weights move only where views exist
- Results depend on the benchmark,  $\tau$ ,  $\Omega$  and the risk aversion
- Without views, Black–Litterman returns the benchmark itself
- Formally a GLS (mixed) estimator: one member of the family of mean-shrinkage estimators with Bayes–Stein and 1/N combinations

## Risk Contributions

- ▣ Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^N w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^N \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- ▣ **Risk contribution** of asset  $i$ :  $RC_i = w_i (\Sigma w)_i / \sigma_p$ ; percentage share  $RC_i / \sigma_p$
- ▣ Equal weights are not equal risks: volatile assets dominate
- ▣ Risk budgeting: choose  $w$  so that the shares  $RC_i / \sigma_p$  equal preset budgets  $b_i$

## Equal Risk Contribution

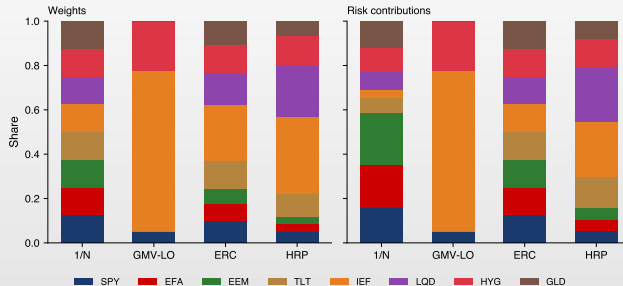
- ERC (Maillard, Roncalli & Teïletche (2010)):  $RC_i = RC_j$  for all  $i, j$
- Computed from a convex problem, then normalised to sum to one

$$\min_{y>0} \frac{1}{2} y^\top \Sigma y - \frac{1}{N} \sum_i \ln y_i, \quad w = y / 1^\top y$$

- Volatility ordering:  $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**,  $w_1 = \sigma_1^{-1} / (\sigma_1^{-1} + \sigma_2^{-1})$ , for any  $\rho \in (-1, 1)$ 
  - SPY (14.7%) and IEF (6.6%): ERC holds 31% SPY; a 60/40 mix lets SPY carry 94% of the risk
- Needs no expected returns, only  $\Sigma$ ; for  $\Sigma$  positive definite a solution exists and is unique (strictly convex objective, infinite at the boundary)

## Weights versus Risk Contributions

Eight multi-asset ETFs, monthly May 2007 - Jul 2026



- Eight multi-asset ETFs, full-sample covariance matrix, May 2007 – July 2026 ( $T = 231$ )

## Reading the Risk Contributions

- ▣ 1/N: equities and high yield receive half of the weight but 69% of the risk
- ▣ ERC: 12.5% of the risk from every asset; IEF gets the largest weight (25%)
- ▣ Long-only GMV: 73% IEF, 22% HYG, 5% SPY; five assets at zero
- ▣ Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%
- ▣ ERC lies between 1/N and GMV, as the theory predicts

## Hierarchical Clustering of Assets

- ▣ Correlation distance:  $d_{ij} = \sqrt{(1 - \rho_{ij})/2} \in [0, 1]$
- ▣ **Single linkage**: repeatedly merge the two closest clusters; cluster distance = closest pair
- ▣ The result is a tree (**dendrogram**); cutting it gives groups of similar assets
- ▣ Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market
- ▣ Textbook: Härdle & Simar (2019), cluster analysis

## Hierarchical Risk Parity

▣ HRP (López de Prado (2016)), three steps

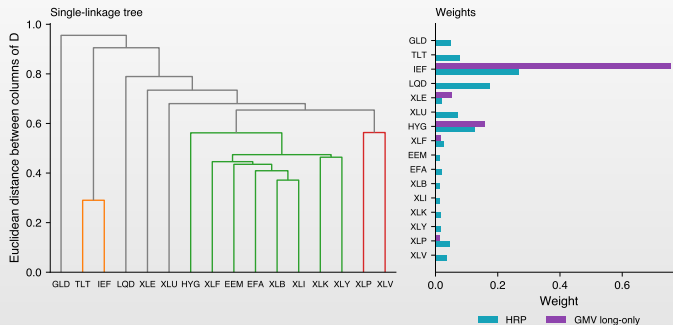
- ▶ Tree clustering: Euclidean distance between the columns of  $D = (d_{ij})$ ,  $\tilde{d}_{ij} = \sqrt{\sum_k (d_{ki} - d_{kj})^2}$ , then single linkage on  $\tilde{d}_{ij}$
- ▶ Quasi-diagonalisation: reorder assets so that similar ones are adjacent
- ▶ Recursive bisection: split the ordered list in two; split the weight in inverse proportion to the variance of each half

▣ Variance of a half: inverse-variance weights inside it,  $\tilde{w}^\top \Sigma \tilde{w}$

▣ No matrix inversion: works even when  $\Sigma$  is singular

▣ All weights are positive

## HRP on 16 ETFs



- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026

## Reading the Tree

- ▣ Closest pair: TLT and IEF ( $\tilde{d} = 0.29$ ); gold joins last (0.96)
- ▣ Equity cluster: cyclical sectors, EFA, EEM and high-yield credit
- ▣ Defensive sectors (utilities, staples, health care) and energy join the equity cluster late
- ▣ HRP: all 16 weights positive, largest IEF 26.7%, LQD 17.6% and HYG 12.6%
- ▣ Long-only GMV: 75.6% in IEF and only five assets above zero

## Other Risk-Based Rules

- ▣ **Inverse volatility:**  $w_i \propto 1/\sigma_i$ ; equals ERC when all correlations are equal
- ▣ **Maximum diversification** (Choueifaty & Coignard (2008)): maximise  $w^\top \sigma / \sqrt{w^\top \Sigma w}$
- ▣ **Minimum variance:** GMV, the extreme risk-based rule
- ▣ All ignore expected returns: robust to estimation error, but optimal only under special assumptions
  - ▶ ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

## Recap: Risk-Based Allocation

- ▣ Risk contributions decompose portfolio volatility additively
- ▣ ERC equalises them; it lies between  $1/N$  and GMV
- ▣ HRP uses a correlation tree and needs no matrix inversion
- ▣ These rules avoid expected returns entirely: less estimation error, no view on returns

## Backtest Design

- ▣ **Out-of-sample** (OOS): weights at month  $t$  use only data up to  $t - 1$
- ▣ Rolling window of 60 months; rebalancing every month
- ▣ Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- ▣ Three universes: sectors (OOS January 2004 – July 2026, 271 months), multi-asset and combined (May 2012 – July 2026, 171 months)
- ▣ Annualised mean, volatility and Sharpe ratio of excess returns; maximum drawdown of total-return wealth

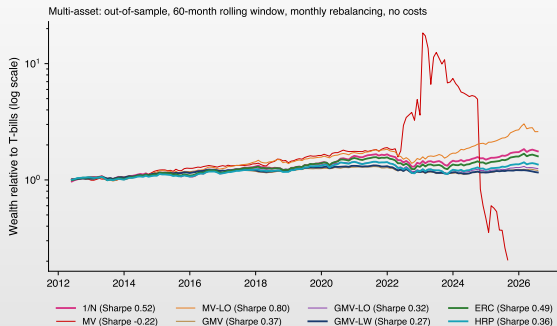
## Turnover and Transaction Costs

- Between rebalancings weights drift with returns

$$w_{i,t}^+ = \frac{w_{i,t}(1 + R_{i,t})}{\sum_j w_{j,t}(1 + R_{j,t})}$$

- Turnover:** traded fraction of wealth,  $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^+|$
- Net return with cost  $c$  per unit traded:  $R_{p,t}^{\text{net}} = R_{p,t} - c TO_t$
- 1/N also trades: it sells winners and buys losers every month

## Out-of-Sample Growth: Multi-Asset

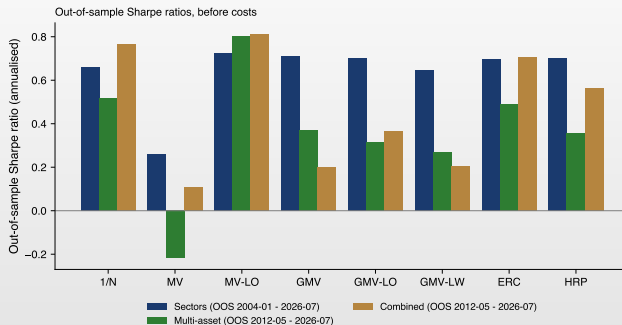


- Eight multi-asset ETFs; wealth relative to a Treasury-bill account, before costs; a path ends at a monthly loss of 100% or more (MV)

## Reading the Multi-Asset Backtest

- ▣ MV collapses: annualised volatility 518%, turnover 44 times wealth per month; a monthly loss above 100% ends its wealth path
- ▣ MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- ▣ GMV rules have the lowest volatility (3.8%–5.4%) but also low Sharpe ratios (0.27–0.37)
  - ▶ They load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT  $-0.5\%$ , IEF  $-0.2\%$  a year)
- ▣ Is MV-LO really better, or lucky? A test is needed

## Sharpe Ratios Across Universes



60-month rolling window, monthly rebalancing, annualised, before costs

MFM\_ch4\_oos\_backtest

## Results: Sector Universe

| Rule   | Mean  | Volatility | Sharpe | Turnover | Max. drawdown |
|--------|-------|------------|--------|----------|---------------|
| 1/N    | 9.4%  | 14.3%      | 0.66   | 0.02     | −49%          |
| MV     | 17.8% | 69.0%      | 0.26   | 3.86     | −100%         |
| MV-LO  | 10.7% | 14.7%      | 0.72   | 0.23     | −42%          |
| GMV    | 8.5%  | 11.9%      | 0.71   | 0.23     | −33%          |
| GMV-LO | 8.1%  | 11.5%      | 0.70   | 0.09     | −34%          |
| GMV-LW | 7.6%  | 11.8%      | 0.65   | 0.14     | −31%          |
| ERC    | 9.3%  | 13.2%      | 0.70   | 0.03     | −47%          |
| HRP    | 8.9%  | 12.8%      | 0.70   | 0.08     | −45%          |

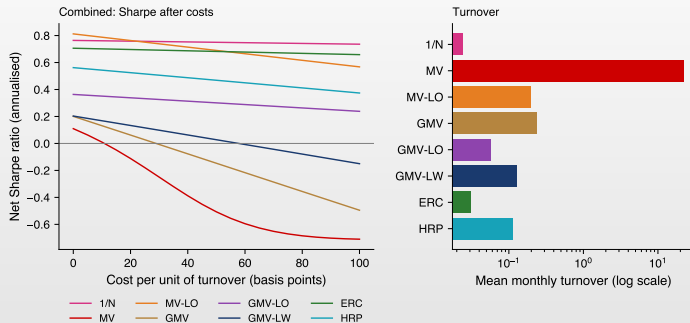
- January 2004 – July 2026; turnover: mean monthly traded fraction of wealth; drawdown of total-return wealth
- MV: a monthly loss above 100% wipes out the capital (drawdown −100%); its mean and Sharpe describe a hypothetical return series, not investable wealth
- All rules except MV land within 0.07 of 1/N

## Results: Combined Universe (16 ETFs)

| Rule   | Mean  | Volatility | Sharpe | Turnover | Sharpe, 50 bp |
|--------|-------|------------|--------|----------|---------------|
| 1/N    | 7.9%  | 10.3%      | 0.76   | 0.02     | 0.75          |
| MV     | 28.7% | 261.8%     | 0.11   | 22.10    | -0.51         |
| MV-LO  | 8.0%  | 9.8%       | 0.81   | 0.20     | 0.69          |
| GMV    | 0.8%  | 4.1%       | 0.20   | 0.24     | -0.15         |
| GMV-LO | 2.0%  | 5.4%       | 0.36   | 0.06     | 0.30          |
| GMV-LW | 0.9%  | 4.4%       | 0.20   | 0.13     | 0.03          |
| ERC    | 5.5%  | 7.8%       | 0.71   | 0.03     | 0.68          |
| HRP    | 4.0%  | 7.0%       | 0.56   | 0.11     | 0.47          |

- May 2012 – July 2026; last column: net of 50 basis points per unit of turnover
- 1 basis point (bp) = 0.01%

## Sharpe Ratios after Costs



Combined universe (16 ETFs); cost  $c$  from 0 to 100 bp per unit of turnover

## Reading the Cost Chart

- ▣ Turnover ranges from 0.02 (1/N) to 22 (MV) per month
- ▣ Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW
- ▣ MV-LO overtakes 1/N only for costs below about 22 bp (Seminar, B8)
- ▣ 1/N and ERC barely move: low turnover is a hidden advantage
- ▣ Rule: report net performance for realistic costs, not only gross

## Inference on One Sharpe Ratio: Lo (2002)

- Lo (2002), i.i.d. Normal returns:  $\sqrt{T}(\widehat{SR} - SR) \rightarrow N(0, 1 + SR^2/2)$ , with  $SR$  monthly
- 1/N on sectors,  $T = 271$ : monthly  $\widehat{SR} = 0.191$ , standard error 0.061
  - ▶ Annualised  $0.66 \pm 1.96 \times 0.21$ : 95% interval  $[0.25, 1.08]$
- Annualising by  $\sqrt{q}$  ( $q = 12$ ) is exact only without autocorrelation; in general  $SR(q) = \eta(q) SR$

$$\eta(q) = \frac{q}{\sqrt{q + 2 \sum_{k=1}^{q-1} (q-k) \rho_k}}$$

- Here  $\rho_1 = -0.03$ ,  $\sum_{k=1}^{11} \rho_k = -0.11$ :  $\eta = 3.62$  instead of  $\sqrt{12} = 3.46$ , annual Sharpe 0.69

## Testing a Difference in Sharpe Ratios

- $H_0$ :  $SR_1 = SR_2$  for two strategies evaluated on the same months
- Jobson & Korkie (1981): a test under i.i.d. Normal returns
- Ledoit & Wolf (2008): a robust version
  - ▶  $\hat{\Delta} = f(\hat{v})$ ,  $v = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$ ,  $f = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$
  - ▶ Standard error by the delta method:  $\sqrt{\nabla f^\top \hat{\Psi} \nabla f / T}$ ,  $\hat{\Psi}$  a HAC covariance
- Strategies on the same assets are highly correlated: small differences can be significant

## The Block Bootstrap



Bradley Efron, inventor of the bootstrap  
(1979)

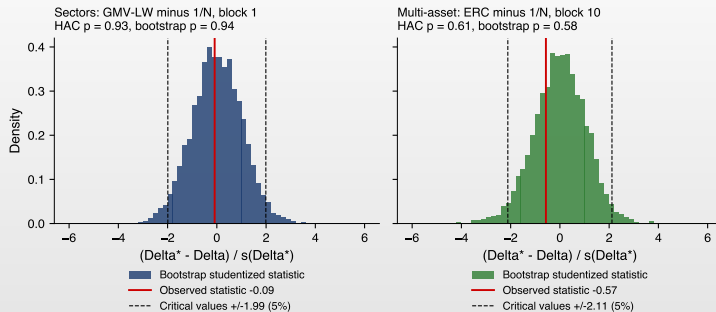
Photo: Ryan K. Morris (2007); public  
domain; Wikimedia Commons

- Bootstrap: resample the data to approximate the sampling distribution of  $\hat{\Delta}$
- Returns are dependent (volatility clustering): resample **blocks** of consecutive months (Künsch (1989))
- Ledoit & Wolf (2008), Section 3.2.2: circular blocks of pairs  $(r_{1t}, r_{2t})$ , wrapping around the end of the sample
- **Studentized**: bootstrap  $|\hat{\Delta}^* - \hat{\Delta}|/s(\hat{\Delta}^*)$ , with  $s(\hat{\Delta}^*)$  from the block sums of each resample
  - ▶ The studentized statistic is asymptotically pivotal: more accurate than the Normal approximation of the HAC test
- $s(\hat{\Delta})$  on the data: prewhitened quadratic-spectral kernel (Andrews & Monahan (1992)), automatic bandwidth (Andrews (1991))
- $p = [\#\{\tilde{d}^{*m} \geq |\hat{\Delta}|/s(\hat{\Delta})\} + 1]/(M + 1)$ ,  $M = 4999$  draws

## Choosing the Block Length: Algorithm 3.1

- Ledoit & Wolf (2008), Algorithm 3.1: choose the block  $b$  whose bootstrap interval has actual coverage closest to 95%
  - ▶ Fit a first-order vector autoregression to the pair of returns; resample its residuals by a stationary bootstrap (Politis & Romano (1994)), mean block 5
  - ▶ Generate  $K = 5000$  pseudo-samples of length  $T$ ; in each, build the 95% interval for  $b \in \{1, 2, 4, 6, 8, 10\}$
  - ▶  $\hat{g}(b)$ : share of intervals that contain the pseudo-true  $\hat{\Delta}$ ; pick  $\tilde{b} = \arg \min_b |\hat{g}(b) - 0.95|$
- The grid and  $K$  are those of the paper (Sections 4–5); applied to each of the 21 pairs of this chapter
- Chosen blocks: mostly 1 for sectors, 8–10 for multi-asset and combined rules (stronger dependence)
- The earlier non-studentized bootstrap with a fixed 6-month block is not the published test and is no longer used

## Is Anything Better than $1/N$ ?



- Bootstrap distributions of the studentized Sharpe ratio difference against  $1/N$ ; red: observed statistic; dashed: bootstrap critical values

## Reading the Tests

- ▣ Sectors, GMV-LW minus  $1/N$ :  $-0.02$ , 95% interval  $[-0.36, 0.33]$ ; HAC  $p = 0.93$ , bootstrap  $p = 0.94$  (block 1)
- ▣ Multi-asset, ERC minus  $1/N$ :  $-0.03$ , interval  $[-0.15, 0.08]$ ; HAC  $p = 0.61$ , bootstrap  $p = 0.58$  (block 10)
  - ▶ ERC and  $1/N$  are so correlated that the interval is narrow, yet it still covers zero
- ▣ Sectors: no rule differs from  $1/N$  at 5%; the closest is ERC,  $+0.04$  (HAC  $p = 0.09$ , bootstrap 0.13)
- ▣ Multi-asset: MV-LO  $+0.29$  is **not** significant (HAC  $p = 0.12$ , bootstrap 0.16)

## Significant Differences: Mostly Losses

| Universe    | Rule minus 1/N | $\hat{\Delta}$ | HAC $p$ | Bootstrap $p$ |
|-------------|----------------|----------------|---------|---------------|
| Multi-asset | MV             | -0.74          | 0.01    | 0.06          |
| Multi-asset | HRP            | -0.16          | 0.07    | 0.05          |
| Combined    | GMV            | -0.57          | 0.03    | 0.05          |
| Combined    | GMV-LO         | -0.40          | 0.05    | 0.08          |
| Combined    | GMV-LW         | -0.56          | 0.02    | 0.03          |
| Combined    | MV-LO          | +0.05          | 0.82    | 0.82          |

- No rule beats 1/N significantly in any universe
- GMV rules lose significantly on the combined universe: low risk, but too little return
- 21 tests: Holm-adjusted  $p$ -values are at least 0.29 (HAC) and 0.68 (bootstrap): nothing survives (Chapter 3, multiple testing)

## Many Rules, One Winner: the Deflated Sharpe Ratio

- ▣ Bailey & López de Prado (2014): the best of  $N$  Sharpe ratios is biased upwards even if every true Sharpe ratio is zero
- ▣ Expected maximum under  $H_0$ : all true Sharpe ratios zero,  $N$  independent trials, monthly units as  $T$ ; with  $\gamma_E = 0.5772$  (Euler's constant) and  $\hat{V}$  the variance of the  $N$  estimates

$$SR_0 = \sqrt{\hat{V}} \left[ (1 - \gamma_E) \Phi^{-1} \left( 1 - \frac{1}{N} \right) + \gamma_E \Phi^{-1} \left( 1 - \frac{1}{Ne} \right) \right]$$

- ▣ Deflated Sharpe ratio, with skewness  $\hat{\gamma}_3$  and kurtosis  $\hat{\gamma}_4$  of the winner

$$DSR = \Phi \left( \frac{(\widehat{SR} - SR_0) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- ▣ This chapter: 24 configurations (8 rules  $\times$  3 universes); winner MV-LO on 16 ETFs, 0.81 a year
  - ▶ As 24 independent trials:  $SR_0 = 0.52$  a year,  $DSR = 0.84 < 0.95$
  - ▶ The configurations are strongly correlated (up to 0.99), so fewer are independent:  $N = 8$  gives DSR 0.93,  $N = 3$  gives 0.98; the verdict hinges on the effective  $N$

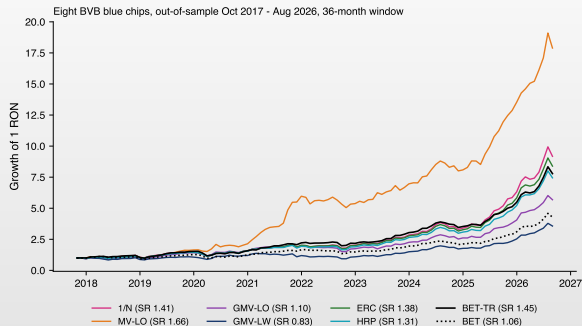
## Recap: Out-of-Sample Evaluation

- ▣ Only out-of-sample, net-of-cost results are informative
- ▣ Sample MV fails in every universe; constraints and shrinkage tame it
- ▣ Risk-based rules and  $1/N$  are hard to separate statistically
- ▣ Test Sharpe differences with HAC errors or the studentized block bootstrap, and count your tests
- ▣ Single Sharpe ratios need standard errors and autocorrelation-aware annualisation; a winner among many needs deflation

## Setup: a Romanian Investor

- ▣ Eight blue chips listed since 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- ▣ Monthly returns in RON, October 2014 – August 2026; 36-month window, first portfolio October 2017
- ▣ Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP; benchmarks BET-TR and BET
- ▣ Sharpe ratios with  $R_f = 0$  (no RON money-market series in the data): return per unit of risk, higher than excess-return Sharpe ratios
- ▣ Earlier Romanian evidence on diversification: [Dragotă, Șerbănescu & Pele \(2008\)](#)

## BVB Portfolios versus BET-TR



□ Out-of-sample, October 2017 – August 2026 (107 months), before costs

## Reading the BVB Backtest

- ▣ BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45; BET: mean 17.8%, Sharpe 1.07
  - ▶ The gap of about 7% a year is the dividends: compare with BET-TR, never with BET
- ▣ 1/N 1.41, ERC 1.38, HRP 1.31: close to BET-TR
- ▣ MV-LO 1.66 (difference +0.21,  $p = 0.44$ ); GMV-LW 0.83 ( $-0.63$ ,  $p = 0.05$ )
- ▣ Average correlation of the eight stocks: 0.44; the latest GMV-LO puts 70% in TLV

## Beyond Equities: Digital Assets

- ▣ [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#): cryptocurrencies in portfolio allocation rules; gains depend on the rule and on liquidity
- ▣ [Pele et al. \(2023\)](#): are cryptos becoming alternative assets?
- ▣ Bitcoin: annualised volatility 172% (monthly, 2010–2026), IEF 6.6%
- ▣ Risk contributions grow with volatility: risk-based rules give Bitcoin a small weight automatically
- ▣ Robust mean–variance for crypto-assets: [Pele \(2023\)](#); Seminar 4, B5

## Recap: BVB Portfolios

- Benchmark choice matters: BET omits about 7% a year of dividends
- No optimised BVB-only rule beats BET-TR significantly
- Eight stocks with average correlation 0.44: little room for diversification inside the BVB
- Open question: does diversification **beyond** BET pay? (Seminar 4, Part C)

## An Open Question

- Does any **estimated** allocation rule beat  $1/N$  out of sample, net of costs, once every rule, window and universe tried is counted?
- Live evidence points both ways
  - ▶ Nonlinear shrinkage of  $\Sigma$  improves GMV out of sample in large cross-sections: [Ledoit & Wolf \(2017\)](#)
  - ▶ Combining tangency, GMV and  $1/N$  under estimation risk: [Kan, Wang & Zhou \(2022\)](#)
  - ▶ Backtests need haircuts for the number of specifications tried: [Harvey & Liu \(2015\)](#)
- Formal statement for  $K$  rules on one universe
  - ▶  $H_0$ :  $\Delta_k = SR_k - SR_{1/N} \leq 0$  for all  $k$ ;  $H_1$ :  $\Delta_k > 0$  for some  $k$
  - ▶ Statistic:  $\hat{\Delta}_k$  with HAC or block-bootstrap errors ([Ledoit & Wolf \(2008\)](#)); error rate fixed for the whole family
- Falsified by: a positive  $\hat{\Delta}_k$  that survives a family-wise correction, costs and a second sample

## The Discovery Loop, Step by Step (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List peer-reviewed studies since 2015 comparing shrinkage or risk-based rules with  $1/N$  out of sample; for each give DOI,  $N$ ,  $T$ , cost assumption and test."*
  - ▶ Resolve every DOI on Crossref and open the table cited; assistants merge authors and invent DOIs
- **Hypotheses:** *"Give three competing explanations for a Sharpe gain over  $1/N$  (better  $\Sigma$ , a low-volatility premium, luck) and the testable implication of each for turnover, subperiods and factor loadings."*
- **Data and code:** *"Write Python that reads data/market, joins prices on common dates and runs a 60-month rolling GMV-LW and  $1/N$ ; weights at  $t$  may use data up to  $t - 1$  only."*
  - ▶ Replicate first: the combined-universe table of this chapter ( $1/N$  0.76, GMV-LW 0.20) before changing anything

## The Discovery Loop, Step by Step (2/2)

- **Robustness:** *"Draft a pre-registration: the family of tests ( $\text{rules} \times \text{universes} \times \text{windows} \times \text{costs}$ ), the error rate (FWER or FDR), the subperiods and a placebo, fixed before any result is seen."*
  - ▶ Placebo: random long-only weights with the same turnover; a real effect must beat their distribution
- **Critique:** *"Act as a referee: give the three strongest reasons my Sharpe gain is spurious and the test that would expose each."*
- **Report:** every specification tried, adjusted  $p$ -values, the power of the design, code and seeds
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

## What the Researcher Must Check

- **Look-ahead:** weights at  $t$  built with month- $t$  returns  $\Rightarrow$  assert that the last estimation date is before  $t$ ; lagging the weights one more month is only a sensitivity check
- **Survivorship and selection:** an ETF list chosen today  $\Rightarrow$  use the universe available at each date, including later delisted ETFs and their final returns; otherwise disclose the limitation
- **Multiple testing:** 7 rules  $\times$  3 universes already give 21 tests  $\Rightarrow$  Holm (FWER) or Benjamini–Hochberg (FDR, [Benjamini & Hochberg \(1995\)](#)) adjusted  $p$ -values
- **Power:**  $\hat{\Delta}$  has a standard error near 0.18 on 171 months  $\Rightarrow$  compute the detectable effect before running
- **Return type:** portfolios aggregate **simple** returns; Sharpe ratios use **excess** returns  $\Rightarrow$  recompute one month step by step
- **Invented references:** an unresolvable DOI or a title mismatch  $\Rightarrow$  drop the claim, log it in `AI_ERRORS.md`

## Mini-Case: Data Snooping in This Chapter's Backtest

- A typical assistant proposal: “MV-LO beats 1/N on the multi-asset universe”;  $H_0: \Delta \leq 0$ , one-sided  $p = 1 - \Phi(\hat{\Delta}/\text{se})$
- Replicate:  $\hat{\Delta} = +0.29$ , HAC standard error 0.18, one-sided  $p = 0.06$  (two-sided HAC 0.12, bootstrap 0.16, Section 7)
- Test the family, not the winner: 21 tests (7 rules  $\times$  3 universes)
  - ▶ Only 7 of 21 differences are positive; the 4 two-sided HAC  $p$ -values below 0.05 are all **losses**, which never reject a superiority null
  - ▶ One-sided: smallest  $p = 0.046$  (sectors, ERC); adjusted Holm 0.96, Benjamini–Hochberg 0.62; 0 discoveries at an FDR of 10%
- Power 0.8 for  $\Delta = 0.29$ , one-sided: about 435 months (36 years) at 5%, 944 months (79 years) at the Holm level 5%/21
- Interpretation: the data cannot rank these rules; the bootstrap interval for MV-LO,  $[-0.13, 0.70]$ , admits losses and large gains; failure to reject is not proof of equality

## Your Turn: A Team-Project Seed

- Question: does nonlinear shrinkage ([Ledoit & Wolf \(2017\)](#)) or a three-fund combination ([Kan, Wang & Zhou \(2022\)](#)) beat  $1/N$  on the 16 ETFs of this chapter, after 10–50 bp costs?
- Replicate first: the combined-universe table (1/N Sharpe 0.76, MV-LO 0.81) and the adjusted  $p$ -values of the mini-case
- Deliverables
  - ▶ A one-page pre-registration: hypotheses, family of tests, error rate, power, placebo
  - ▶ Code on data/market, a table of all specifications, a referee-style self-critique
- Log every prompt, tool and accepted output in `AI_USE.md`; every error caught (invented DOI, look-ahead, wrong return type) and the test that caught it in `AI_ERRORS.md`
- Code of the mini-case: [Quantlets/Ch\\_04/ai\\_discovery\\_case.py](#)

## Conclusions

- ▣ Markowitz is the right question with fragile answers: estimation error dominates
- ▣ Sample MV needs decades of data to beat  $1/N$ ; out of sample it fails in every universe
- ▣ Shrinkage, constraints, Black–Litterman and risk-based rules make portfolios stable
- ▣ No rule beats  $1/N$  significantly here; several lose after costs
- ▣ Evaluate out of sample, net of costs, with a test of the Sharpe difference

## Links to Other Chapters

- ▣ **Chapter 3** (factor models): the tangency portfolio and the CAPM; factor covariance models
- ▣ **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- ▣ **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- ▣ **Chapter 8**: backtesting without overfitting
- ▣ **Chapter 13**: machine learning for returns and portfolios

## Further Study on Quantinar

### Advanced courses on quantinar.com

- ▣  Hierarchical Clustering
- ▣  Minimum Spanning Tree
- ▣  Multivariate Statistical Analysis

### Code and data

- ▣ All charts and tables: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_04](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_04)
- ▣ Risk-free rate: Kenneth French Data Library

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